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Deep feature optimization for enhanced fish freshness assessment

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ABSTRACT

Assessing fish eye freshness is vital for ensuring food safety and minimizing economic losses in the seafood industry. However, traditional sensory evaluation methods remain subjective, time-consuming, and inconsistent. Despite recent advancements in deep learning for automating visual freshness prediction, challenges related to accuracy and feature transparency persist. This study introduces a unified three-stage framework that refines and leverages deep visual representations for reliable fish eye freshness assessment. First, five state-of-the-art vision architectures – ResNet-50, DenseNet121, EfficientNet-B0, ConvNeXt-Base, and Swin-Tiny – are fine-tuned to establish a strong baseline. Next, multi-level deep features extracted from these backbones are used to train seven classical machine learning classifiers, integrating deep and traditional decision mechanisms. Finally, feature selection methods based on Light Gradient Boosting Machine (LGBM), Random Forest, and Lasso are utilized to identify a compact and informative subset of features. Experiments conducted on the Freshness of the Fish Eyes (FFE) dataset demonstrate that the best configuration – combining Swin-Tiny features, a Random Forest classifier, and LGBM-based feature selection – achieves an accuracy of 85.99%, outperforming recent studies on the same dataset by 8.69–22.78%. These findings confirm the effectiveness and generalizability of the proposed framework for visual quality evaluation tasks.

1. Introduction

Fish is an essential component of the global diet, providing high-quality protein and vital nutrients that contribute significantly to human health (Tidwell and Allan, 2001; Chen et al., 2022). Beyond its nutritional value, the fisheries sector also plays a crucial economic role in global food security and trade. However, fish is an inherently perishable commodity, and improper handling or storage can rapidly deteriorate its freshness, leading to both economic losses and potential health risks. Traditionally, freshness evaluation has relied on sensory inspection by trained experts, who assess the eyes, gills, skin, and odor. Although practical, these subjective methods are time-consuming, inconsistent, and difficult to scale for large quantities of fish, particularly under industrial conditions (Hassoun and Karoui, 2017; Prabhakar et al., 2020). Therefore, developing rapid, accurate, and automated techniques for assessing fish eye freshness has become essential to ensure food safety and improve management efficiency in the seafood industry.

To address these challenges, researchers have explored a wide range of image-based approaches for objective and automated evaluation. Early studies primarily employed handcrafted visual descriptors – such as color features, local binary patterns (LBP), and gray-level co-occurrence matrices (GLCM) – combined with machine learning classifiers to infer freshness levels (Medeiros et al., 2021; Syarwani et al., 2022; Arora et al., 2022; Hoang et al., 2026). While these methods achieved promising results, their performance heavily depended on manual feature design and often lacked robustness across diverse imaging conditions. This limitation has motivated the adoption of deep learning approaches that can automatically learn hierarchical and discriminative representations directly from raw images.

The advent of deep learning has revolutionized visual analysis by enabling automatic feature learning directly from raw images. Several convolutional neural network (CNN)-based architectures and lightweight variants have been applied to fish eye freshness prediction (Prasetyo et al., 2022b; Jayasundara et al., 2023; Cahyo and Al-Ghiffary, 2024; Khaleel et al., 2024; Sanga et al., 2024). However, despite clear progress, the Freshness of the Fish Eyes (FFE) dataset remains particularly challenging due to its high variability involving eight fish species across three freshness levels. Furthermore, the images were captured using mobile phones under uncontrolled lighting and background conditions, which explains why existing deep models have achieved only moderate accuracies of 63% and 77% in the studies of Prasetyo et al. (2022b), Yildiz et al. (2024). This suggests that distinguishing subtle inter-class differences in eye appearance still demands more discriminative and interpretable representations.

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Moreover, existing studies often evaluate deep networks in an end-to-end manner, without explicitly analyzing how feature abstraction levels or feature selection mechanisms affect performance. For instance, Prasetyo et al. (2022b) directly used CNN or pre-trained models for classification without optimizing or interpreting the extracted embeddings, while Yildiz et al. (2024) focused solely on feature extraction from CNNs without systematic dimensionality refinement.

In this context, this study introduces a comprehensive deep feature optimization framework that systematically evaluates and refines learned representations for fish eye freshness assessment. Specifically, we:

- Evaluate several state-of-the-art vision backbones – including ResNet-50, DenseNet-121, EfficientNet-B0, Swin-Tiny, and ConvNeXt-Base – to establish a robust baseline for image-based freshness classification.
- Apply Grad-CAM visualization to interpret and compare model behavior across architectures, without guiding the framework design.
- Extract deep embeddings from different abstraction levels of each backbone and evaluate their discriminative power using classical machine learning classifiers.
- Investigate embedded feature selection methods – LGBM (boosting), Random Forest (bagging), and Lasso (L1 regularization) – to identify compact and discriminative subsets of features.

To the best of our knowledge, this is the first study to systematically analyze how deep feature abstraction and embedded feature optimization jointly affect fish eye freshness classification performance on the public FFE dataset. The proposed framework demonstrates improved accuracy and enhanced interpretability, while maintaining computational efficiency suitable for practical applications.

The rest of the paper is organized as follows. Section 2 reviews related literature. Section 3 presents the proposed methodology, covering the main approaches and techniques used. Section 4 describes the experimental setup. Section 5 presents the results and offers a comprehensive discussion of the findings. Finally, Section 6 concludes the study and suggests directions for future research.

2. Related works

Early computer vision approaches for fish eye freshness assessment primarily relied on manually engineered features extracted from digital images to capture visual degradation in the eyes, gills, and skin. Color-based descriptors were dominant, often computed as statistical measures or histograms in RGB, HSV, HSI, and CIELAB spaces to quantify discoloration. For instance, Medeiros et al. (2021) extracted a comprehensive set of colorimetric parameters from multiple color spaces and achieved 100% accuracy in classifying tuna and salmon freshness using an AutoML framework. Texture features like the GLCM and LBP were also used to represent surface variations such as eye cloudiness. Syarwani et al. (2022) combined HSV color features with GLCM descriptors, attaining 94.28% accuracy for Nile Tilapia freshness using a SVM. Arora et al. (2022) further advanced feature fusion by computing a unified “Q-score” from weighted visual features of the gills, eyes, and skin, achieving 98.07% accuracy. More recently, Hoang et al. (2026) demonstrated that a systematic fusion of handcrafted color and texture features could achieve 77.56% accuracy on the FFE dataset, highlighting the potential of optimized feature integration for improving machine learning-based fish eye freshness classification.

Building on these foundations, deep learning approaches were developed to extract discriminative features automatically. Lightweight architectures such as MobileNetV1 bottleneck with Expansion (MB-BE) achieved 63.21% accuracy on the FFE dataset, illustrating the trade-off between efficiency and performance (Prasetyo et al., 2022b). Jayasundara et al. (2023) developed FishNET-S for Indian Sardinella focusing on fish eyes and FishNET-T for Yellowfin Tuna focusing on fish meat, achieving 84.1% and 68.3% accuracy, respectively. Concurrently, many studies focused on well-established CNN architectures such as ResNet and DenseNet. By leveraging pre-trained models and data fusion techniques, these approaches reported exceptional classification accuracies ranging from approximately 93% to 100% (Cahyo and Al-Ghiffary, 2024; Khaleel et al., 2024; Sanga et al., 2024).

Recent research has explored advanced hybrid models that combine multiple neural network architectures to improve fish eye freshness assessment. Rodrigues et al. (2024) were the first to apply Vision transformers (ViT) in this context. Their two-stage system first segmented the fish eye region with high performance, achieving a 98.77% detection rate and 85.7% IoU using a Segformer, before classification with a ViT resulted in 80.8% accuracy. Hybrid architectures have also been explored, with Biswas et al. (2025) proposing a CNN–LSTM model that integrates spatial and sequential information and uses LIME for interpretability, achieving 86% accuracy. Peries et al. (2025) conducted a systematic comparison of multiple pre-trained architectures across whole fish, fish eyes, and fish gills, with the highest accuracy of 99.13% achieved on gills using DenseNet121.

Hybrid frameworks have become an important direction in automated fish eye freshness assessment, employing deep learning to generate rich feature representations and traditional machine learning to perform classification. This combination leverages the strengths of both approaches, balancing accuracy with efficiency and interpretability. For example, Kılıçarslan et al. (2024) extracted features from pre-trained models and classified them with traditional algorithms, achieving a 100% success rate. Similarly, Yildiz et al. (2024) applied pre-trained CNNs to the FFE dataset and found that combining VGG19 features with an ANN yielded the highest accuracy of 77.3%. Extending this concept, Lanjewar and Panchbhai (2024) employed a NasNet–LSTM architecture for feature extraction along with data balancing and feature selection techniques, achieving Matthew’s correlation coefficient (MCC) and Cohen’s kappa coefficient (KC) scores of 99.1%.

Another advanced strategy is multimodal data fusion, which integrates information from different sensors to provide a more comprehensive representation of freshness. Hardy et al. (2024) fused fluorescence, visible, and near-infrared spectroscopy data, reporting nearly perfect accuracies of 99.5% compared to single-mode analyses of 77.1%. Similarly, Balım et al. (2025) combined RGB images with laser reflectance data, achieving 88.44% accuracy.

A review of the literature also highlights several common limitations, as summarized in Table 1. Many studies rely on small or private datasets collected under ideal laboratory conditions, such as controlled lighting and background, which hinders fair comparison and raises concerns about generalizability. Methodologically, much of the research focuses on a single anatomical region such as the eye or relies exclusively on either handcrafted features or deep learning, rarely exploring a systematic combination of both. Furthermore, some hybrid frameworks use conventional CNN architectures that may not fully capture complex visual cues.

These limitations highlight the need for a systematic framework that not only evaluates deep feature representations across different abstraction levels but also incorporates feature optimization techniques to achieve compact, interpretable, and generalizable freshness classification.

3. Methodology

3.1. Overview of the proposed approach

To systematically evaluate and overcome the inherent difficulty of accurately classifying fish eye freshness from imaging data, this

Table 1
Summary of recent studies on fish eye freshness assessment.

Studies	Methodology	Accuracy	Limitations
Medeiros et al. (2021)	Colorimetric features from multiple spaces classified with AutoML	100%	Limited number of tuna and salmon samples. Freshness was assessed only using color features, which may overlook other important indicators
Syarwani et al. (2022)	HSV + GLCM features with SVM classifier	94.28%	Small sample of Nile Tilapia eyes and gills in lab conditions
Arora et al. (2022)	Fused gill, eye, and skin features into a single Q-score	98.07%	Limited Rohu data reduces generalizability across species and environments. Segmentation employed simple traditional image processing
Prasetyo et al. (2022b)	Proposed a lightweight CNN (MB-BE)	63.21%	Low overall accuracy due to lightweight design and limited feature representation and limited generalization
Jayasundara et al. (2023)	Proposed two specialized CNNs: FishNET-S (for small fish eyes) and FishNET-T (for large fish meat)	90% (FishNET-S), 100% (FishNET-T)	Two separate models for different fish species limit generalization and controlled data reduce robustness in real-world settings
Cahyo and Al-Ghiffary (2024)	ResNet101 + GLCM features	100%	The research focused on a single species with limited data, employing conventional image processing for segmentation in controlled lab lighting
Khaleel et al. (2024), Sanga et al. (2024)	ResNet18 + image fusion; Compared VGG19, MobileNetV2, DenseNet201, ResNet50	93%, 100%	Considering only two freshness classes, the deep learning model's interpretability remains limited
Rodrigues et al. (2024)	Segformer for segmentation + ViT for classification	98.77% detection, 85.7% IoU, 80.8% accuracy	Using transformers for segmentation and classification requires high computational resources
Biswas et al. (2025)	CNN–LSTM integrating spatial–sequential features with LIME interpretability	86%	The integration of CNN and LSTM on a single-species dataset with binary classification increases model complexity
Peries et al. (2025)	Evaluated CNNs and pre-trained models on fish eyes, gills, and whole fish	99.13% (DenseNet121, gill)	Low image resolution, binary classification, single-region data, and limited explainability reduce generalization
Kılıçarslan et al. (2024)	Features from MobileNetV2, Xception, VGG16 + ML classifiers (SVM, LR, ANN, RF)	100%	Single-species dataset under controlled conditions, binary classification, and limited explainability
Yildiz et al. (2024)	VGG19 & SqueezeNet features + ML classifiers (KNN, RF, SVM, LR, ANN)	77.3%	Extracted features are insufficient to capture complex, long-range dependencies in the data
Lanjewar and Panchbhai (2024)	NasNet–LSTM hybrid with SMOTEENN balancing and feature selection	99.1% (MCC & KC scores)	Focusing only on the eye region and using a CNN–LSTM hybrid, the model becomes more complex and harder to interpret
Hardy et al. (2024)	Fused fluorescence + VIS–NIR data via LDA, KNN, and ensemble bagged trees	99.5%	Limited fish samples, generous classification metric, controlled lab environment, and single-species focus
Balım et al. (2025)	Laser + CNN feature fusion with SVM, MLP, RF classifiers	88.44%	The evaluation used only mackerel and a 940 nm laser wavelength, making generalization to other species and conditions difficult

study proposes a novel and progressive framework that integrates modern deep learning and machine learning techniques within a unified training–evaluation pipeline (Fig. 1). The framework is designed not merely as a comparative setup but as a structured exploration of how recent advances in vision architectures and hybrid learning strategies can be effectively exploited for fine-grained classification tasks.

In the first stage, a series of representative deep learning models – ResNet-50, DenseNet-121, EfficientNet-B0, Swin-Tiny, and ConvNeXt-Base – were fine-tuned using Random search for hyperparameter optimization. These architectures were deliberately chosen to capture diverse design philosophies, ranging from conventional convolutional networks to transformer-based and convolution–transformer hybrid

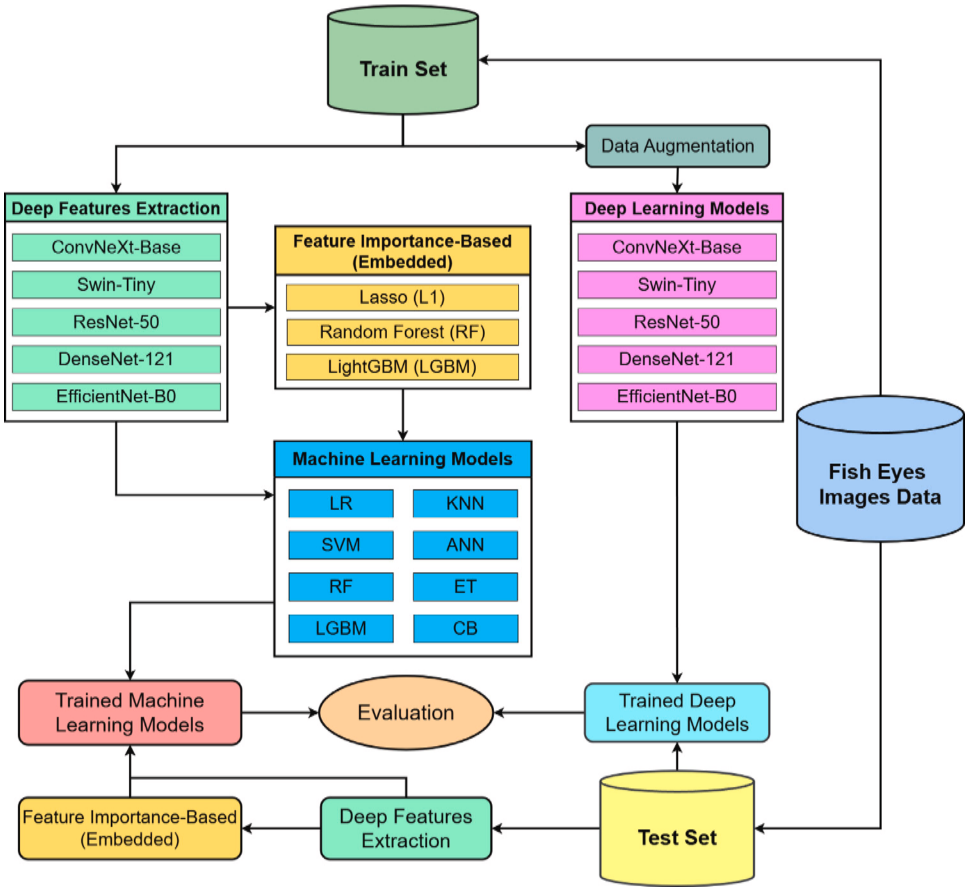


Fig. 1. Overview of the proposed methodology.

models. This phase aimed to establish a robust baseline and to investigate how architectural diversity and the Random search strategy influence the representation of visual freshness cues.

In the second stage, the study transitions to a hybrid deep feature-based learning framework. Rather than relying solely on fully abstract representations, deep features were extracted from the intermediate-to-high abstraction levels of each fine-tuned backbone, specifically from the global average pooling (GAP) outputs. Instead of combining these representations, the discriminative power of features from each stage was individually evaluated to determine which level of abstraction yields superior classification performance. This design choice was motivated by the characteristics of fish-eye images, where freshness cues depend on both fine-grained visual details (e.g., texture and glossiness) and higher-level semantic attributes (e.g., overall color consistency and shape). Evaluating both levels helped identify the most informative representation that balances local texture information with global semantic cues, making it particularly suitable for this application. Two complementary analysis pathways were then explored:

(1) Full deep features, where all extracted representations were directly used for classification; and

(2) Feature-selected deep features, where redundant or less informative dimensions were removed using embedded feature selection methods–LASSO (L1 regularization), RF (bagging), and LGBM (boosting)–to automatically rank and retain the most discriminative subset of features.

This design enabled a systematic comparison between the raw representational capacity and the compact, feature-selected variants, highlighting the trade-offs between richness and efficiency of learned embeddings.

The resulting feature sets (either full or feature-selected) were then used as inputs to a suite of classical machine learning algorithms, including Logistic Regression (LR), K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest (RF), Extra Trees (ET), LightGBM (LGBM), and CatBoost (CB). This stage examined whether explicit decision mechanisms could further enhance discriminative power beyond the softmax-based classification of deep networks.

In the final evaluation stage, both fine-tuned deep models and ML classifiers trained on deep features were assessed under identical conditions using standard performance metrics. Grad-CAM visualization was employed to interpret spatial attention in deep models, offering a comprehensive view of model behavior and interpretability.

The following section presents a detailed description of the methods used in this study.

3.2. Deep learning models

To evaluate the performance and efficiency of different deep learning architectures, this study employs a set of representative models that reflect diverse design paradigms in visual recognition. The selected architectures comprise classical convolutional neural networks (ResNet-50 and DenseNet-121), a lightweight and efficiency-oriented model (EfficientNet-B0), a transformer-based architecture (Swin Transformer-Tiny), and a modern convolutional network inspired by transformer design principles (ConvNeXt-Base). This architectural diversity enables a fair and systematic comparison of how different inductive biases influence fish eye freshness assessment.

3.2.1. ResNet-50

ResNet (He et al., 2016) introduces residual learning to address the optimization challenges of very deep neural networks. Instead of learning direct mappings, it employs shortcut connections that enable residual functions, improving gradient flow and convergence. The

ResNet-50 architecture consists of 50 layers arranged in four residual stages with downsampling at the start of stages 2, 3, and 4. Each bottleneck block includes 1×1, 3×3, and 1×1 convolutions, while the network begins with a 7×7 convolution and max pooling, and concludes with global average pooling and a fully connected classification layer.

3.2.2. DenseNet-121

Dense Convolutional Networks (DenseNets) (Huang et al., 2017) introduce dense connectivity to enhance feature reuse and parameter efficiency. Instead of summing residuals as in ResNet, each layer in a DenseNet concatenates its output with all preceding feature maps within the same block. DenseNet-121 applies this principle across multiple dense blocks separated by transition layers that downsample and reduce channels using 1×1 convolutions and average pooling. Each dense layer consists of batch normalization, ReLU activation, and convolution, followed by global average pooling and a fully connected classification layer.

3.2.3. EfficientNet-B0

EfficientNet (Tan and Le, 2020) introduces a compound scaling method that uniformly balances network depth, width, and input resolution using learned coefficients, addressing the inefficiency of scaling a single dimension. EfficientNet-B0 serves as the baseline architecture discovered through neural architecture search, optimized for both accuracy and computational cost. Its primary building block, the MBConv, employs an inverted bottleneck with channel expansion, depthwise separable convolution, and Squeeze-and-Excitation attention. The model follows a standard multi-stage ConvNet structure, concluding with global average pooling and a fully connected classification layer.

3.2.4. Swin Transformer-Tiny (Swin-Tiny)

The Swin Transformer (Liu et al., 2021) introduces a hierarchical Vision Transformer that improves efficiency for high-resolution and dense prediction tasks. It builds hierarchical feature maps through patch merging and applies window-based self-attention within local regions for linear computational complexity. The Swin Transformer-Tiny (Swin-T) variant uses shifted windows to capture cross-window dependencies, with patch merging between stages to reduce tokens and expand feature dimensions before global pooling and classification.

3.2.5. ConvNeXt-Base

ConvNeXt (Liu et al., 2022) revisits conventional ConvNet architectures by incorporating design principles from Vision Transformers to achieve comparable performance while maintaining convolutional efficiency and inductive biases. ConvNeXt-Base follows a ResNet-like stage structure with key modifications, including a larger-stride stem, non-overlapping patch-like convolutions, large-kernel depthwise layers, and inverted bottlenecks inspired by Transformer MLPs. It replaces batch normalization with layer normalization and applies downsampling between stages, forming a hierarchical yet purely convolutional architecture.

3.3. Deep feature extraction strategy

The effectiveness of deep features hinges on a crucial trade-off: mid-level layers retain local, fine-grained spatial details necessary for subtle cues, such as texture and glossiness, while high-level layers capture richer, more abstract semantic patterns. To systematically identify the most informative representations for assessing fish eye freshness, features from both mid-level and high-level stages of each fine-tuned model were independently extracted. These representations were subsequently condensed using GAP, a standard technique that creates compact, fixed-length vectors from feature maps. This approach preserves essential global information while significantly reducing feature dimensionality and computational complexity for the downstream traditional machine learning models. The specific candidate extraction points for each model, along with their detailed rationale, are summarized in Table 2.

3.4. Explainability with grad-CAM

Grad-CAM (Gradient-weighted Class Activation Mapping) is a visualization technique used to interpret the decision-making process of deep learning models by highlighting the important regions in the input image that contribute most to the model's prediction (Selvaraju et al., 2017). Grad-CAM computes the gradients of the target class score with respect to the feature maps of the last convolutional layer (in CNNs) or the corresponding representation layer (in Transformers) to generate a heatmap indicating the areas of focus.

In this study, Grad-CAM is applied to both CNN- and Transformer-based models to analyze the visual regions influencing their classification decisions. For CNN architectures (ResNet-50, DenseNet-121, EfficientNet-B0, and ConvNeXt-Base), Grad-CAM is computed using the feature maps from the final convolutional layer, which preserves high-level semantic information while retaining spatial structure.

For the Transformer-based model (Swin-Tiny), Grad-CAM requires a specific adaptation due to its hierarchical and non-convolutional design. In this case, we use the output of the final Swin Transformer block in the last stage (Stage 4), where discriminative semantic representations are formed prior to classification. Gradients of the target class score are backpropagated with respect to the patch-level feature representations at this stage.

As the Swin Transformer produces feature maps in a channel-last format (B, H, W, C), where B, H, W, and C denote the batch size, feature map height, feature map width, and number of channels, respectively, a dimension permutation is applied to convert them into the standard channel-first format (B, C, H, W) before computing the class activation maps, ensuring a consistent Grad-CAM formulation across CNN and Transformer architectures.

3.5. Embedded feature selection strategy

In this study, we adopt embedded feature selection as our primary approach. Embedded methods integrate the assessment of feature relevance directly into the model training process, unlike filter and wrapper techniques, which treat feature selection as a separate preprocessing step. By evaluating predictive value during model construction (Fig. 2), embedded methods produce a compact, model-specific subset of features that enhances both robustness and generalization. Our choice of embedded feature selection is motivated by prior evidence demonstrating its effectiveness across diverse domains. For example, in water quality index (WQI) prediction (Lap et al., 2023) and rice seed purity classification (Phan and Nguyen, 2025), embedded techniques consistently outperformed filter and wrapper approaches in terms of predictive performance and feature selection capability. Considering the high dimensionality and complexity of deep features extracted for fish eye freshness assessment, embedded methods are expected to provide a reliable and efficient strategy for selecting the most informative features.

To perform comprehensive feature optimization, we employ three complementary embedded methods: Lasso regression (L1) - Regularization), RF, and LGBM. These techniques cover the major mechanisms of embedded feature selection:

1. Tree-based ensemble selection (Bagging & Boosting)

- **Random forest (Bagging):** RF averages importance across many independent decision trees. Features are ranked based on their ability to improve node purity (e.g., Gini impurity), providing a stable and generalized measure of relevance.
- **LightGBM (Boosting):** LGBM sequentially builds trees, focusing on features that provide the greatest Gain (reduction in loss) at each split. This effectively captures complex non-linear relationships and the predictive power of features.

Table 2
Candidate deep feature extraction points for comparative analysis.

Model	Mid-level extraction	Rationale	High-level extraction	Rationale
ResNet-50	Layer 3 (1024-dim)	Captures mid-level patterns before final downsampling	Layer 4 (2048-dim)	Encodes abstract semantic features
DenseNet-121	DenseBlock 3 (1024-dim)	Rich fused representation from interconnected layers	DenseBlock 4 (1024-dim)	Final refined feature set before output
EfficientNet-B0	Block 6 (192-dim)	Captures complex spatial patterns with larger receptive field	Block 7 (320-dim)	Refines features for high-level representation
Swin-Tiny	Stage 3 (384-dim)	Contextual features from self-attention layers	Stage 4 (768-dim)	Captures global representation of image tokens
ConvNeXt-Base	Stage 3 (512-dim)	Aggregated features from deep ConvNeXt blocks	Stage 4 (1024-dim)	Highest-level feature hierarchy

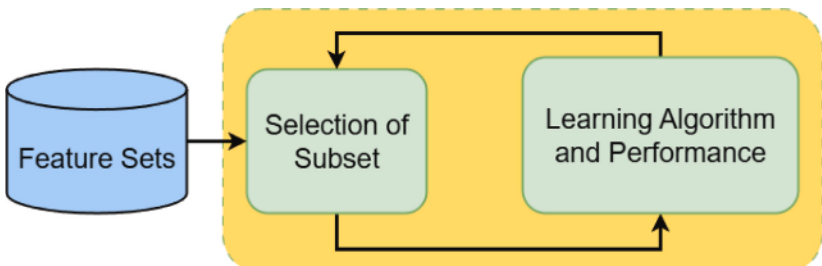


Fig. 2. Mechanism of embedded feature selection in model training.

2. Regularization-based selection (Lasso regression (L1)): Lasso introduces a penalty proportional to the absolute magnitude of coefficients. The L1 penalty drives coefficients of less informative features to exactly zero, effectively performing selection. This is particularly effective for managing multicollinearity in high-dimensional feature spaces.

By employing three embedded feature selection methods – Lasso, RF, and LGBM – our approach ensures that the selected feature subsets are optimized, stable, and minimally redundant for fish eye freshness classification. For each method, the selection threshold was empirically determined by testing multiple cutoffs based on feature importance scores or coefficient magnitudes.

3.6. Machine learning models

This study investigates the classification of fish into three freshness categories ("*Highly Fresh*", "*Fresh*", "*Not Fresh*") using a diverse set of machine learning algorithms. The models were selected based on their ability to handle structured data, computational efficiency, and capability to capture both linear and complex non-linear relationships. A detailed description of each model is presented below.

Logistic Regression (LR): is a statistical model that uses a logistic function to predict the probability of a binary outcome (Hosmer and Lemeshow, 2000). It transforms a linear combination of input features into a value between 0 and 1. The model is trained by finding the coefficients that maximize the likelihood of the observed data, and these coefficients indicate the influence of each feature on the predicted probability.

K-Nearest Neighbors (KNN): is a simple yet effective non-parametric algorithm for classification and regression (Fix and Hodges, 1951). It classifies a sample by identifying its k closest neighbors based on a distance metric such as Euclidean or Manhattan distance and assigning the majority label. By relying on local data structure, KNN captures patterns without assuming any specific underlying distribution.

Support Vector Machines (SVM): is a supervised learning algorithm that identifies the optimal hyperplane separating classes with maximum margin (Cortes and Vapnik, 1995). The closest data points, known as support vectors, define this boundary. For non-linear relationships, kernel functions project data into higher-dimensional spaces, enabling effective classification of complex patterns.

Artificial Neural Networks (ANN): are computational models inspired by biological neural systems (McCulloch and Pitts, 1943). They consist of interconnected layers of neurons that transform inputs through weighted connections and non-linear activations. Trained via backpropagation to minimize prediction error, ANNs can model complex non-linear relationships, enabling effective performance across classification and regression tasks.

Random Forest (RF): is an ensemble learning algorithm that builds a collection of decision trees to improve classification performance (Breiman, 2001). Each tree is trained on a random subset of the data and a random subset of features, which introduces diversity and reduces overfitting. To classify a new sample, each tree makes a prediction, and the final class is determined by majority voting among all trees. This approach allows Random Forest to capture complex patterns and interactions in the data while maintaining high accuracy and robustness.

Extra Trees (ET): is an ensemble learning algorithm similar to Random Forest that constructs a collection of decision trees (Geurts et al., 2006). Unlike Random Forest, Extra Trees select split points completely at random for each feature when building the trees, which increases variability and reduces overfitting. Each tree provides a class prediction, and the final output is determined by majority voting among all trees. This method allows Extra Trees to capture complex patterns efficiently while being fast and robust.

LightGBM (LGBM): is a gradient boosting framework that constructs decision trees sequentially to minimize residual errors (Ke et al., 2017). It incorporates Gradient-based One-Side Sampling (GOSS) to emphasize informative samples and Exclusive Feature Bundling (EFB) to reduce feature dimensionality. Designed for efficiency and scalability, LGBM delivers fast and accurate predictions on large datasets with mixed feature types.

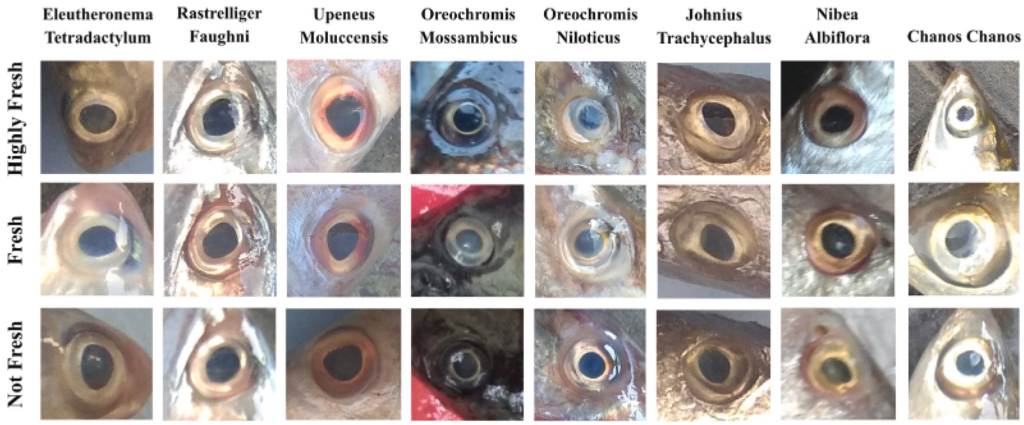


Fig. 3. Sample images from the *Freshness of the Fish Eyes (FFE)* dataset representing eight fish species and three freshness categories (*Highly Fresh*, *Fresh*, and *Not Fresh*).

Table 3
Distribution of images in the FFE dataset across the training, validation, and testing subsets.

Freshness class	Training set	Validation set	Testing set
Highly Fresh	1148	250	366
Fresh	838	211	271
Not Fresh	823	242	241

CatBoost (CB): is a gradient boosting algorithm that constructs an ensemble of decision trees sequentially to minimize prediction errors (Dorogush et al., 2018). It efficiently handles categorical features through ordered boosting, which mitigates overfitting and enhances generalization. By combining the outputs of all trees, CatBoost achieves high accuracy on complex datasets with both numerical and categorical variables.

4. Experiments

4.1. Data description

The dataset used in this study is the publicly available *Freshness of the Fish Eyes (FFE)* collection introduced by Prasetyo et al. (2022b). It was specifically designed for classifying fish eye freshness based on visual cues from the eyes. The dataset contains a total of 4390 images categorized into three freshness levels: 1764 "*Highly fresh*" (days 1–2), 1320 "*Fresh*" (days 3–4), and 1306 "*Not fresh*" (days 5–6) (see Fig. 3 for sample images).

Images were captured daily over a six-day period using a mobile phone under varied backgrounds and lighting conditions to simulate real-world scenarios. This ensures that the dataset reflects practical variations encountered in typical fish markets. The provided images have been pre-processed to focus on the eye region, which serves as the primary visual input for the models. For model development and evaluation, the dataset was split into three subsets: 64% for training, 16% for validation, and 20% for testing, using stratified and consistent splits across all experiments with a random seed of 42 (see Table 3).

4.2. Experimental setup

All experiments in this study were conducted on two computational platforms. A local machine, equipped with an AMD Ryzen 7 processor, 16 GB of RAM, and an NVIDIA RTX 3050 GPU, was used for model development and classical machine learning algorithms. Computationally intensive deep learning tasks were carried out on the Kaggle platform, which provides an NVIDIA Tesla T4 GPU with 16 GB of VRAM.

Table 4
Experimental settings for deep learning and classical models.

Aspect	Configuration
Image preprocessing	Resize: 224 × 224, Normalization: ImageNet stats
Data augmentation	Horizontal flip ($p = 0.5$), Rotation: ±30°, Brightness jitter (0.2)
DL initialization	ImageNet pre-trained weights
Training setup	Optimizer: AdamW, Learning rate: 1e−4, Weight decay: 1e−2, Batch size: 64
Epochs	Max: 100, Early stopping patience: 20
Feature normalization	StandardScaler
ML hyperparameters	Random search optimization
Evaluation	Identical conditions for all models

The preprocessing procedures, training configurations, and optimization strategies applied in all experiments are summarized in Table 4. This table provides a comprehensive overview of the experimental settings to ensure reproducibility and facilitate fair comparison across models.

For the classical machine learning models, hyperparameters were tuned using random search within the parameter ranges summarized in Table 5.

5. Results and discussion

The proposed methodologies for fish eye freshness assessment were evaluated through five state-of-the-art deep learning models and hybrid frameworks that combine deep representations with classical classifiers. Furthermore, the influence of feature selection on model performance was analyzed. To ensure a thorough and balanced evaluation, four widely recognized metrics were employed: Accuracy (overall correctness), Precision (positive predictive value), Recall (sensitivity), and F1-Score (harmonic mean of Precision and Recall). Together, these metrics provide a comprehensive measure of both general and class-specific performance.

5.1. Performance of deep learning models

The experimental results of five state-of-the-art deep learning architectures revealed their strong discriminative capability in fish eye freshness assessment. As presented in Table 6, all models achieved accuracies above 80%, confirming the robustness of deep neural networks in handling this challenging visual classification task.

Table 5
Hyperparameter search space for classical machine learning models used in the experiments.

Model	Hyperparameter	Search Space/Values
Logistic Regression (LR)	penalty	l1, l2
	C	0.01, 0.1, 1.0, 10, 100
	solver	liblinear, saga
	max_iter	Integer range [300, 799]
K-Nearest Neighbors (KNN)	n_neighbors	3, 5, 7, 9, 11, 15, 17, 19, 21, 23
	weights	uniform, distance
	metric	euclidean, manhattan, chebyshev, minkowski
	algorithm	auto, ball_tree, kd_tree, brute
Support Vector Machine (SVM)	kernel	linear, poly, rbf
	C	0.01, 0.1, 1, 10, 100
	gamma	scale, auto
	coef0	0.0, 0.01, 0.1 (poly kernel only)
Random Forest (RF)	n_estimators	Integer range [10, 500]
	max_depth	Integer range [2, 49]
	criterion	gini, entropy
Extra Trees (ET)	n_estimators	Integer range [10, 500]
	criterion	gini, entropy
	max_depth	None, 10, 15, 20, 25, 30, 35, 40, 50
	min_samples_split	2, 3, 5, 7, 9, 11
	min_samples_leaf	1, 3, 5, 8, 9, 11
	bootstrap	True, False
	max_leaf_nodes	None, 2, 3, 4, 5, 6, 8, 9, 10, 11, 13
LightGBM (LGBM)	boosting_type	gbdt, dart
	n_estimators	Integer range [10, 500]
	learning_rate	0.001, 0.01, 0.1, 1, 10
	num_leaves	Integer range [15, 49]
	max_depth	Integer range [2, 49]
	class_weight	balanced, None
CatBoost (CB)	iterations	Integer range [300, 999]
	learning_rate	0.03, 0.04, 0.05, 0.1, 0.2, 0.3
	l2_leaf_reg	1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21
	grow_policy	SymmetricTree, Depthwise
Neural Network (ANN)	hidden_layer_sizes	Integer range [64, 256] (single hidden layer)
	activation	logistic, tanh, relu
	solver	adam, sgd, lbfgs
	alpha	0.0001, 0.001, 0.01, 0.1
	max_iter	Integer range [800, 1700]
	learning_rate	constant, adaptive, invscaling
	early_stopping	True

Among all evaluated models, the Swin Transformer-Tiny (Swin-Tiny) achieved the highest accuracy of 84.85%, outperforming the other architectures. This superior performance can be attributed to its hierarchical transformer design with window-based self-attention, which effectively captures both fine-grained local textures (e.g., lens cloudiness and surface opacity) and broader contextual cues (e.g., gradual color transitions and spatial consistency) relevant to fish-eye freshness. Unlike conventional convolutional models with fixed receptive fields, Swin-Tiny enables more effective modeling of these discriminative visual characteristics through localized self-attention and shifted windows, thereby contributing to its improved classification performance.

The ConvNeXt-Base model reached a closely similar accuracy of 84.51%. This strong performance can be attributed to its modernized convolutional design, which incorporates principles inspired by vision transformers. Features such as larger kernel sizes, updated block structures, and enhanced normalization strategies allow ConvNeXt to capture both fine-grained textures and larger spatial patterns effectively, resulting in performance comparable to transformer-based models.

EfficientNet-B0, DenseNet-121, and ResNet-50 are considered mid-level models, achieving accuracies of 81.32%, 80.75%, and 80.07%, respectively. EfficientNet-B0 leverages compound scaling to uniformly adjust depth, width, and resolution, capturing rich visual features efficiently. Its MBConv blocks with squeeze-and-excitation further enhance feature representation by emphasizing informative channels. DenseNet-121 and ResNet-50 rely on conventional architectures that effectively extract general features but may be limited in modeling complex spatial relationships and subtle textures, which are important for fine-grained freshness assessment.

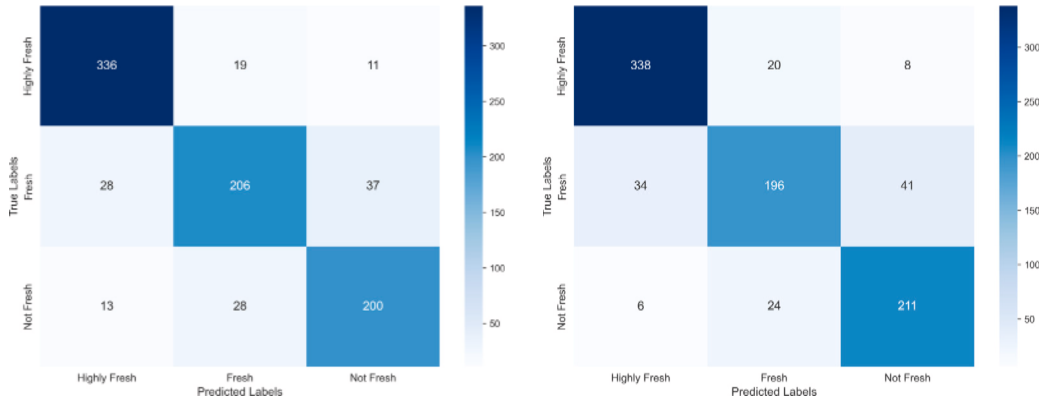
Table 6
Performance of different DL models for fish eye freshness assessment (%).

Model	Accuracy	Precision	Recall	F1-score
ConvNeXt-Base	**84.51**	83.73	83.60	83.62
Swin-Tiny	**84.85**	**84.08**	**84.08**	**83.93**
EfficientNet-B0	81.32	80.55	80.04	80.19
DenseNet-121	80.75	79.91	80.04	79.97
ResNet-50	80.07	79.21	79.59	79.33

5.1.1. Confusion matrix analysis

To gain a clearer understanding of the specific inter-class misclassifications made by the models, we provide a detailed analysis of the Confusion Matrices (CM) for the two best-performing architectures: ConvNeXt-Base and Swin-Tiny, as shown in Fig. 4.

The analysis revealed that both models were highly effective at identifying the extreme classes. Swin-Tiny accurately classified 338 "*Highly fresh*" samples, while ConvNeXt-Base performed comparably well by correctly identifying 336 samples. However, the models diverged in handling the critical boundary zones. ConvNeXt-Base was marginally better at identifying the difficult middle class, correctly classifying 206 "*Fresh*" samples compared to Swin-Tiny's 196. At the same time, it demonstrated a weakness in risk assessment, as it misclassified 13 truly "*Not fresh*" samples as "*Highly fresh*". In contrast, Swin-Tiny was more



(a) ConvNeXt-Base (b) Swin-Tiny

Fig. 4. Confusion matrices of ConvNeXt-Base and Swin-Tiny for fish eye freshness assessment.

robust against this critical failure, making the same error on only 6 samples. This suggests that while ConvNeXt-Base is slightly better at the subtle distinctions within the "*Fresh*" category, Swin-Tiny provides superior reliability because it is far less likely to over-evaluate spoiled fish, which is a key requirement for safety-critical applications.

Furthermore, the observed misclassification patterns do not appear to be driven by class imbalance, as the FFE dataset exhibits a relatively balanced class distribution. Instead, most confusions occur between adjacent freshness categories, suggesting that they arise from subtle and continuous visual changes in fish eye degradation. This interpretation is further supported by the Grad-CAM analysis presented in the following section (Section 5.1.2) and the ambiguous samples illustrated in Fig. 9, which highlight how overlapping visual characteristics – such as lens opacity – can lead to challenging class boundaries.

5.1.2. Grad-CAM analysis

To better understand how the deep learning models make decisions, we applied Grad-CAM, which highlights the specific regions in the images that most influence the models' predictions. The heatmaps shown in Fig. 5 provide important insights into both the internal reasoning of the models and the quality of the feature representations they learned. In particular, the highlighted regions correspond well with known biological and visual indicators of fish eye freshness, such as changes in eye lens transparency, pupil clarity, and surrounding color consistency, rather than irrelevant background regions. This correspondence suggests that the models focus on biologically meaningful cues instead of exploiting spurious correlations.

For the top-performing models, Swin-Tiny and ConvNeXt-Base, the Grad-CAM results are particularly informative. Swin-Tiny shows attention to both fine details of the central pupil and the larger structure of the eye. This ability to integrate local and global information allows the model to form a comprehensive understanding of fish eye freshness. ConvNeXt-Base also demonstrated focused attention, though slightly more diffuse, balancing detailed feature extraction with broader contextual understanding. This precise and balanced focus indicates that these models captured meaningful, discriminative features rather than random patterns, which contributes to their high accuracy. In comparison, other models show clear limitations in their attention patterns. EfficientNet-B0 focused almost entirely on the pupil, which highlighted local textural changes but limited its awareness of surrounding structures, likely contributing to the high adjacency misclassifications. DenseNet-121 spread its attention more widely due to its dense connectivity. While this encouraged feature reuse, it appeared to dilute the focus on the most informative regions. ResNet-50 showed a scattered attention map with substantial background influence, suggesting difficulties in accurately localizing key areas.

These visualizations confirm the strength of transformer-based features, which we further exploit in hybrid setups below.

5.2. Performance of full deep features

After evaluating the strong performance of deep learning architectures in Section 5.1, we further investigated whether the discriminative representations learned by these backbones could be effectively exploited by classical machine learning algorithms, instead of relying on the original fully connected layers with a softmax output. By replacing the neural classification head with alternative ML classifiers, this stage aims to determine whether such algorithms can match or even surpass the discriminative capability of the learned deep models under identical experimental conditions. Consequently, a hybrid learning strategy was adopted: feature representations were extracted from the middle-to-high layers of two major convolutional or transformer blocks of each fine-tuned backbone (e.g., ConvNeXt, Swin-Tiny, EfficientNet-B0, ResNet-50, and DenseNet-121), thereby capturing complementary levels of abstraction. These feature maps were then processed using Global Average Pooling (GAP) to obtain compact, fixed-length embeddings, which subsequently served as inputs to a suite of traditional machine learning classifiers, including LR, KNN, SVM, ANN, RF, ET, LGBM, and CatBoost. All experiments in this section were conducted under the same holdout protocol (64% for training, 16% for validation, and 20% for testing). This unified setup ensures a fair and consistent comparison between the DL models and the proposed hybrid feature-based approach.

Based on the previous analysis, the Swin-Tiny model, which achieved the best overall performance among the evaluated architectures, was selected as the primary backbone for feature extraction. To further investigate the effect of feature abstraction, both middle-level and high-level embeddings were evaluated using identical machine learning classifiers. As shown in Table 7, features extracted from higher layers consistently yielded better results across all classifiers. While the best performance at the middle level was achieved by SVM with an accuracy of 84.97%, all models benefited from the higher-level representations, where ET achieved the highest accuracy of 85.88%, marginally outperforming the Swin-Tiny model (84.85%). This observation suggests that deeper representations are more discriminative, capturing richer semantic information that is crucial for distinguishing subtle variations in fish eye freshness. In contrast, middle-level features – although containing finer textural details – may lack the semantic consistency required for optimal classification.

In addition, we analyzed the confusion matrix of the best-performing configuration (Stage 4 features + ET). As shown in the confusion matrix (Fig. 6), this combination achieved balanced and consistent performance across all three freshness categories. The model correctly identified most samples in each class, with accuracy rates of 91.5% for Highly fresh, 78.9% for Fresh, and 85.1% for Not fresh,

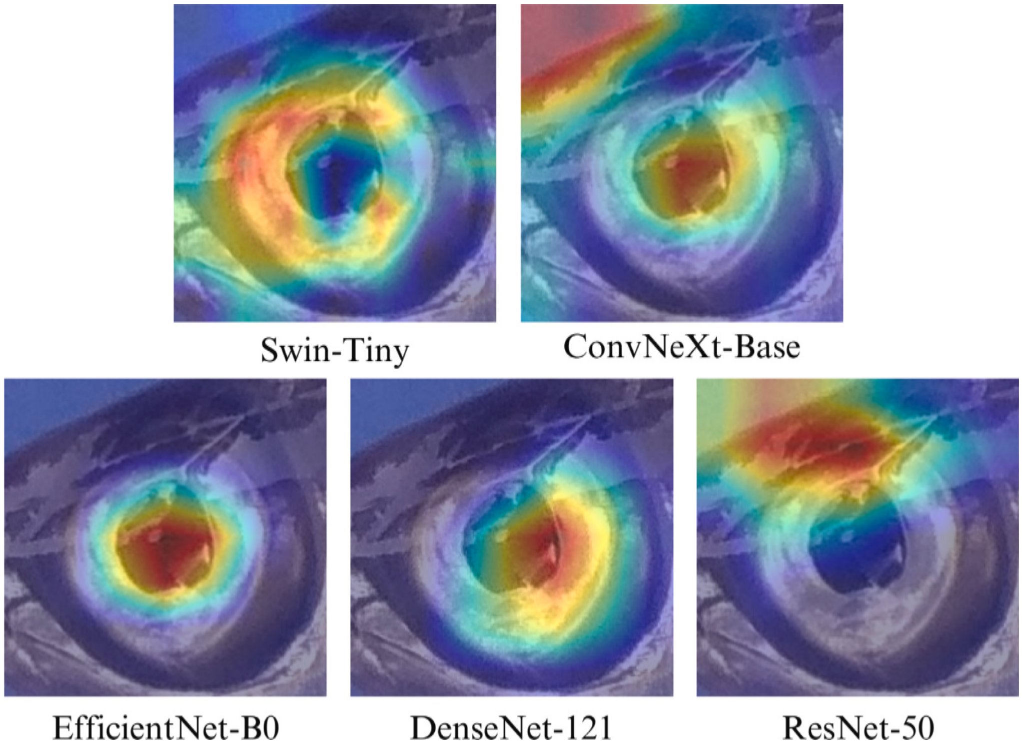


Fig. 5. Grad-CAM visualizations highlighting the image regions attended to by different deep learning models during fish eye freshness classification.

Table 7
Performance of ML classifiers on deep features extracted from Swin-Tiny (Stage 3 vs. Stage 4) (%).

Model	Stage 3 (middle-level features)				Stage 4 (high-level features)			
	ACC	Recall	Precision	F1	ACC	Recall	Precision	F1
LR	81.89	81.89	81.87	81.82	85.19	85.19	85.34	85.24
KNN	83.03	81.03	83.12	83.10	85.65	85.65	85.67	85.64
SVM	**84.97**	84.97	84.99	**84.98**	85.65	85.65	85.58	85.59
ANN	82.23	82.23	82.36	82.26	85.54	85.54	85.67	85.57
RF	79.95	79.95	79.75	79.69	85.31	85.31	85.48	85.38
ET	80.75	80.75	80.52	80.43	**85.88**	**85.88**	**85.91**	**85.89**
LGBM	81.66	81.66	81.55	81.59	85.08	85.08	85.15	85.10
CB	80.64	80.64	80.50	80.54	84.28	84.28	84.35	84.30

resulting in an overall accuracy of 85.88%. Compared to the Swin-Tiny model, this approach improved the classification of the Fresh class by approximately 6.6%, indicating that the tree-based classifier can better handle subtle variations between adjacent categories. Although a slight decrease was observed for the Not Fresh class, the results remain competitive and more evenly distributed. These findings suggest that the Stage 4 embeddings of Swin-Tiny provide sufficiently discriminative representations, and when combined with ET, they enhance generalization performance.

Following the selection of the best-performing backbone, we also conducted experiments using feature embeddings extracted from all other deep architectures to verify the generality of this hybrid approach. The same feature-extraction and training protocol was consistently applied across ConvNeXt, EfficientNet, ResNet-50, and DenseNet to ensure comparability. Importantly, the improvement observed with Swin-Tiny was not an isolated case. As shown in Table 8, the hybrid deep feature-based learning strategy consistently enhanced performance across all evaluated architectures. For instance, ConvNeXt-Base improved from 84.51% to 85.19%, while more traditional CNNs showed even greater gains—ResNet-50 increased from 80.07% to 83.14%, and DenseNet-121 from 80.75% to 83.49%. These results confirm that the fully connected layers in DL models may not always provide the most effective decision boundaries, and that classical machine learning classifiers can more efficiently exploit the rich representations generated by deep backbones.

A crucial observation from Table 8 is that, across all evaluated architectures, the most effective features were consistently extracted from the higher-level representation layers. This pattern reinforces the earlier finding from Swin-Tiny that deeper embeddings provide superior discriminative power compared to those from intermediate levels. ET achieved the best results with transformer-derived representations extracted from Swin-Tiny (768 features) and ConvNeXt-Base (1024 features), highlighting its capacity to model complex, high-level dependencies. KNN performed effectively with DenseNet-121 features (1024 features), consistent with DenseNet's feature-reuse mechanism that forms compact and locally coherent clusters in the embedding space. LGBM showed strong performance with ResNet-50 features (2048 features), demonstrating its robustness in capturing non-linear relationships within high-dimensional data. Finally, SVM achieved the best results with EfficientNet-B0 features (320 features), as its compound scaling and MBConv design produce compact and regularized feature spaces that enhance margin-based separation.

5.3. Performance of selected deep features

The analysis in Section 5.2 demonstrated that high-level features extracted from deep architectures, when used with classical machine

Table 8
Maximum accuracy achieved by each deep learning model using features from its optimal architectural level (%).

Deep feature	Best classifier	Feature source	Dimension	Best Acc	Impact
ConvNeXt-Base	ET	Stage 4 (Final stage)	1024	85.19	+0.68%
Swin-Tiny	ET	Stage 4 (Final stage)	768	**85.88**	+1.03%
EfficientNet-B0	SVM	Block 7	320	82.69	+1.37%
DenseNet-121	KNN	DenseBlock 4	1024	83.49	+2.74%
ResNet-50	LGBM	Layer 4	2048	83.14	+3.07%

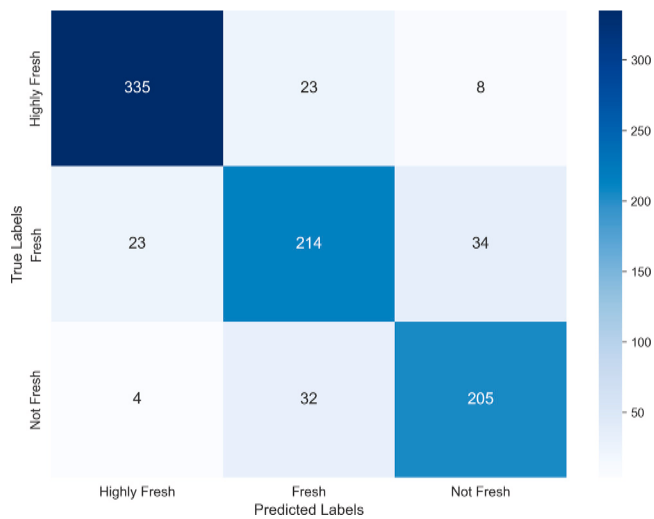


Fig. 6. Confusion matrix of ET using features extracted from Stage 4 of Swin-Tiny.

Table 9
Impact of feature selection using LGBM (boosting) (%).

Feature subset	Dimension	Best classifier	Accuracy	Impact
100% (Full set)	768	ET	85.88	Baseline
50%	384	ET	**85.99**	+0.11%
40%	308	KNN	85.65	−0.23%
30%	231	ET	**85.99**	+0.11%
20%	154	SVM	85.88	+0.00%
10%	77	RF	**85.99**	+0.11%
5%	39	RF	85.54	−0.34%

Table 10
Impact of feature selection using RF (bagging) (%).

Feature subset	Dimension	Best classifier	Accuracy	Impact
100% (Full set)	768	ET	85.88	Baseline
50%	384	SVM	85.42	−0.46%
40%	308	KNN	85.65	−0.23%
30%	231	KNN	85.88	0%
20%	154	RF	85.42	−0.46%
10%	77	SVM	85.54	−0.34%
5%	39	KNN	85.42	−0.46

Table 11
Impact of feature selection using Lasso Regression (L1) (%).

Feature subset	Dimension	Best classifier	Accuracy	Impact
100% (Full set)	768	RF	85.88	Baseline
50%	384	SVM	85.54	−0.34%
40%	308	SVM	85.54	−0.34%
30%	231	KNN	85.42	−0.46%
20%	154	ET	85.31	−0.57%
10%	77	SVM	85.76	−0.12%
5%	39	KNN	84.97	−0.94%

learning classifiers, are highly discriminative. However, their high dimensionality (320–2048 features) can introduce noise, redundancy, and low-predictive-power features, increasing computational cost and potentially limiting performance. This section investigates whether a smaller, more informative subset of features can improve overall performance. Three embedded feature selection methods – LGBM (boosting), RF (bagging), and Lasso regression (L1) – were applied to filter irrelevant predictors. The impact of dimensionality reduction was examined stepwise by progressively retaining the top 50%, 40%, 30%, 20%, 10%, and 5% of features. This gradual reduction enabled us to observe performance trends and identify the optimal trade-off between feature subset size and predictive accuracy before a noticeable drop occurred. The analysis focuses on features extracted from the Swin-Tiny backbone, which achieved the best performance in previous experiments, to evaluate the impact of feature selection.

5.3.1. Impact of feature selection using LGBM (boosting)

The feature selection process with LGBM, a boosting-based method, produced particularly informative results, as shown in Table 9. Reducing the feature dimension did not degrade performance; the highest accuracy of 85.99% was achieved in three cases: RF with 77 features, ET with 231 features, and ET with 384 features. Among these, RF with only 77 features represents the most compact subset while retaining maximum discriminative power, yielding a +0.11% improvement over the baseline accuracy of 85.88%. This corresponds to a 90% reduction in features, highlighting the efficiency of a small, informative subset. These results demonstrate that LGBM effectively filters out redundant or noisy features, leading to a more robust and computationally efficient model.

5.3.2. Impact of feature selection using RF (bagging)

Feature selection using RF (bagging) did not provide any performance gains, as shown in Table 10. While reducing the number of features did not substantially harm classification accuracy, no subset outperformed the full feature set. The peak accuracy of 85.88% was achieved with the top 30% of features (231 dimensions), matching the baseline. Other subsets led to minor decreases in accuracy, ranging from −0.23% to −0.46%. These findings indicate that, despite the strong generalization capability of RF, its feature importance estimation is less capable than LGBM's in identifying a concise set of discriminative features for the FFE dataset.

5.3.3. Impact of feature selection using Lasso regression (L1)

Lasso regression (L1), a regularization-based method, was less suitable for this task, as shown in Table 11. Unlike the tree-based methods, L1 consistently reduced classification accuracy. Performance generally declined as the number of retained features decreased. Even the largest subset, the top 50%, resulted in a −0.34% drop. Notably, with the top 10% of features (77 dimensions), the decrease was smaller (−0.12%), suggesting that Lasso can preserve some discriminative information in very compact subsets. Overall, these results indicate that the relationships between deep features and fish eye freshness classes are highly complex and non-linear, and Lasso's linear penalization likely removed features crucial for capturing these patterns, which are better handled by non-linear classifiers such as ET and SVM.

We can observe that LGBM consistently yields the most effective feature subsets, which can be attributed to its distinctive algorithmic

Table 12
Best-performing feature subsets selected by LGBM (boosting) for each deep learning model (%).

Model	Feature subset	Dimension	Best classifier	Accuracy
Swin-Tiny	10%	77	RF	**85.99**
ConvNeXt-Base	10%	103	SVM	85.31
DenseNet-121	10%	103	ET	83.60
ResNet-50	10%	205	LGBM	83.49
EfficientNet-B0	50%	160	KNN	82.35

properties. As a gradient boosting-based method, LGBM performs feature selection implicitly through gradient-based optimization, where feature importance is determined by the cumulative contribution of each feature to loss reduction across all boosting iterations rather than by isolated splits. Its histogram-based splitting strategy enables efficient evaluation of candidate thresholds, while the leaf-wise tree growth prioritizes highly discriminative splits, allowing LGBM to capture complex nonlinear relationships and subtle feature interactions that are critical for the target task. In contrast, RF evaluates feature importance mainly based on impurity or variance reduction across independently trained trees, which can bias the selection toward features with higher variance or more potential split points and may retain redundant information. Meanwhile, L1 regularization performs feature selection within a linear modeling framework by shrinking coefficients toward zero, which limits its ability to model nonlinear patterns or interactions among features. As a result, LGBM tends to select a feature subset that is both compact and highly informative, effectively filtering out noisy or redundant features while preserving strong predictive cues, as reflected by the improved classification accuracy achieved with a substantially reduced feature set.

5.3.4. Effect of LGBM feature selection across deep architectures

To provide a broader perspective, Table 12 shows the best-performing feature subsets obtained through LGBM-based selection across all evaluated deep architectures. A consistent trend emerges: for most models – Swin-Tiny, ConvNeXt-Base, DenseNet-121, and ResNet-50 – the top 10% of selected features achieved the highest accuracy, outperforming the full feature sets. For instance, ConvNeXt-Base attained its best result (85.31%) with an SVM classifier using 103 features, indicating that LGBM effectively preserves nonlinear discriminative information. DenseNet-121 performed best with ET (83.60%), while ResNet-50 reached 83.49% with LGBM, reflecting robustness in moderately high-dimensional spaces.

In contrast, EfficientNet-B0 deviates from the general trend, as its classification performance slightly decreased after feature selection (from 82.69% to 82.35%). This behavior can be attributed to its architecture, which is explicitly optimized for feature efficiency. Unlike ConvNeXt-Base or ResNet-50, which produce high-dimensional embeddings (1024 and 2048 features, respectively) with substantial redundancy, EfficientNet-B0 generates a compact 320-dimensional representation using compound scaling and MBConv blocks. As a result, the original feature space already contains limited noisy or irrelevant information. Further reducing this representation to 160 dimensions may therefore remove complementary discriminative cues, leading to a slight performance degradation. This observation suggests that post-hoc feature selection is more effective for architectures with redundant representations, whereas compact backbones such as EfficientNet-B0 benefit less from aggressive dimensionality reduction.

5.4. Computational complexity and efficiency analysis

5.4.1. Analysis of model parameters and training time

In addition to predictive accuracy, this study conducted an analysis of model complexity and computational cost, as illustrated in Fig. 7, to emphasize the trade-offs between different architectures. The findings indicate that parameter count is not the sole factor influencing computational expense; rather, architectural design plays a critical role in overall efficiency. To quantify this, we computed parameter efficiency as parameters (in millions) divided by training time per epoch (in seconds), which yielded units of M/s to measure processing speed.

The comparison highlights the contrast between complexity and efficiency. ConvNeXt-Base (87.6M parameters, 89.0 s per epoch, efficiency ≈ 0.98 M/s) exemplifies a highly complex design that achieves strong accuracy but demands considerable resources, making it resource-intensive and potentially limiting its practicality for deployment on constrained hardware. In contrast, EfficientNet-B0 (4.1M parameters, 29.3 s per epoch, efficiency ≈ 0.14 M/s) represents the opposite extreme, offering remarkable efficiency at the cost of reduced accuracy.

The classic architectures provide additional insight into these dynamics. DenseNet-121 (7.6M parameters, 39.1 s per epoch, efficiency ≈ 0.19 M/s) has far fewer parameters than ResNet-50 (23.6M parameters, 36.4 s per epoch, efficiency ≈ 0.65 M/s), yet it requires longer training time due to its feature concatenation operations. This outcome underscores that architectural choices, such as DenseNet's dense connectivity, can outweigh raw parameter efficiency in terms of computational overhead.

Within this context, Swin-Tiny emerged as the most balanced solution. With 27.5M parameters and a training time of 44.1 s per epoch (efficiency ≈ 0.62 M/s), it represents a moderate resource investment while achieving the highest accuracy among all models. It represented fewer computational resources than ConvNeXt-Base (about 31% fewer parameters and 50% less time per epoch), while still outperforming EfficientNet-B0 and DenseNet-121 in both accuracy and efficiency (e.g., 4.4 times higher than EfficientNet-B0's 0.14 M/s), with only a modest increase in cost compared to ResNet-50. This combination of leading accuracy and manageable resource requirements identifies Swin-Tiny as the most compelling backbone for developing models that are both high-performing and practically deployable.

Based on these observations, the analysis highlighted that model selection is a multi-dimensional decision involving more than predictive accuracy. While complex models like ConvNeXt-Base can achieve high accuracy and lightweight models like EfficientNet-B0 offer efficiency, neither extreme is optimal on its own. A balanced architecture such as Swin-Tiny provided a practical compromise, delivering state-of-the-art accuracy with manageable computational cost—particularly beneficial for applications in resource-limited environments like edge computing devices. This makes it suitable for further hybrid model development in real-world scenarios.

5.4.2. Analysis of inference time

Following the analysis of model parameters and training time, inference speed is a critical factor for evaluating the practical feasibility of a model, particularly for applications requiring rapid response or deployment on resource-constrained devices. In this study, we also analyze the inference time per image of the best-performing hybrid framework, which consists of deep feature extraction from Stage 4 of Swin-Tiny, followed by RF-based classification using the feature subset selected by LGBM. The detailed inference time results are reported in Table 13, which indicates that the total inference time per image is 17.878 ms. Notably, most of this time—specifically 17.865 ms, accounting for more than 99.9%–is spent on the deep feature extraction stage using the Swin-Tiny backbone. In contrast, the inference time of the RF classifier is only 0.013 ms, which is negligible. This result highlights the efficiency of the proposed hybrid approach. The feature selection using LGBM, which is performed offline during training, reduces the feature dimensionality from 768 to 77 compact features, enabling the RF classifier to produce predictions almost instantaneously.

The inference-time analysis demonstrates that the proposed hybrid framework effectively balances strong feature representation and fast prediction, making it suitable for real-time deployment in automated aquatic product quality monitoring systems.

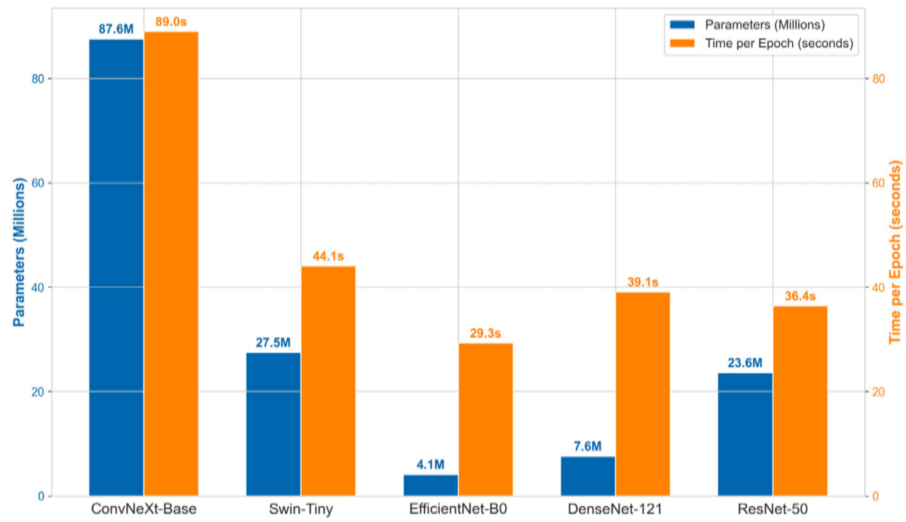


Fig. 7. Comparison of model complexity and training time per epoch across vision architectures.

Table 13
Total inference time per image of the best-performing hybrid framework.

Component	Time per image (ms)
Deep feature extraction (Swin-Tiny)	17.865
ML classifier inference (RF)	0.013
Total (end-to-end)	**17.878**

5.5. Comparison with existing studies using the same dataset

To emphasize the contributions of this study, a comparative analysis was conducted against previously published works using the same *Freshness of the Fish Eyes (FFE)* dataset. As illustrated in Fig. 8, the proposed methods demonstrate clear improvements in both classification accuracy and robustness compared with existing approaches.

The best-performing model, the proposed hybrid method (features from Swin-Tiny + RF), achieved 85.99% accuracy, surpassing the results of Hoang et al. (2026) by 8.43%, Yildiz et al. (2024) by 8.69%, and Prasetyo et al. (2022b) by 22.78%. Even the baseline deep learning model (Swin-Tiny) attained 84.85% accuracy, exceeding all previous results. These outcomes highlight the advantage of using a modern vision transformer backbone and the benefit of combining complementary feature representations.

The superiority of the proposed framework stems from its ability to address the main limitations of earlier studies. The approach by Hoang et al. (2026) is limited by its reliance on handcrafted features, which requires significant domain expertise for engineering and may fail to capture the abstract visual cues that deep learning models learn automatically. Similarly, the hybrid model proposed by Yildiz et al. (2024) was constrained by its use of conventional convolutional backbones (VGG19 and SqueezeNet), limiting its capacity to model long-range dependencies. In contrast, the lightweight MB-BE architecture of Prasetyo et al. (2022b) emphasized computational efficiency at the expense of representational capacity. Its compact design likely constrained the ability to learn complex, fine-grained visual cues, particularly given the limited size and variability of the training dataset.

The proposed framework achieves higher predictive accuracy while providing a balanced and generalizable representation of visual information. It captures both semantic patterns and physical cues, offering a reliable foundation for future research and practical applications in automated fish eye freshness assessment.

5.6. Discussion

The proposed hybrid framework substantially improves the accuracy of fish eye freshness assessment on the FFE dataset, achieving 85.99% using the Swin-Tiny backbone combined with a RF classifier and LGBM-based feature selection. This notable improvement highlights the effectiveness of integrating deep visual representations with classical ML and embedded feature optimization. The three-stage design plays a crucial role in this success: fine-tuning captures rich semantic features; traditional classifiers define sharper and more flexible decision boundaries than a softmax layer; and feature selection isolates compact, informative subsets that reduce noise and enhance generalization.

Regarding generalizability, although the experimental evaluation was conducted on the FFE dataset, the proposed framework itself is not inherently restricted to a specific dataset, fish species, or imaging setup. This interpretation is supported by the consistent performance improvements observed across multiple backbone architectures, including both CNN-based and transformer-based models, as well as different classical classifiers throughout the experimental analysis. The combination of deep feature learning, embedded feature selection, and lightweight classification constitutes a generic methodological design that can be transferred to other fish eye freshness datasets and broader food-quality imaging tasks. Nevertheless, the magnitude of performance improvements is expected to depend on dataset scale, image acquisition conditions, and domain-specific characteristics.

From a practical perspective, the proposed framework offers several advantages for real-world fish eye freshness assessment and related food-quality inspection tasks. By decoupling deep feature extraction from the final decision-making stage, the framework enables the use of lightweight and interpretable classifiers while retaining the discriminative power of deep representations. This design reduces computational complexity and facilitates flexible deployment across different imaging setups and hardware constraints. Moreover, the integration of embedded feature selection contributes to compact feature representations, which can improve efficiency and robustness when scaling to larger datasets or resource-limited environments. In addition, the best-performing configuration achieved an average inference time of 17.878 ms per image, indicating that the proposed framework can operate efficiently in near-real-time scenarios. These properties make the proposed approach particularly suitable as a decision-support tool in automated or semi-automated quality control pipelines, where consistency, adaptability, and computational efficiency are critical considerations.

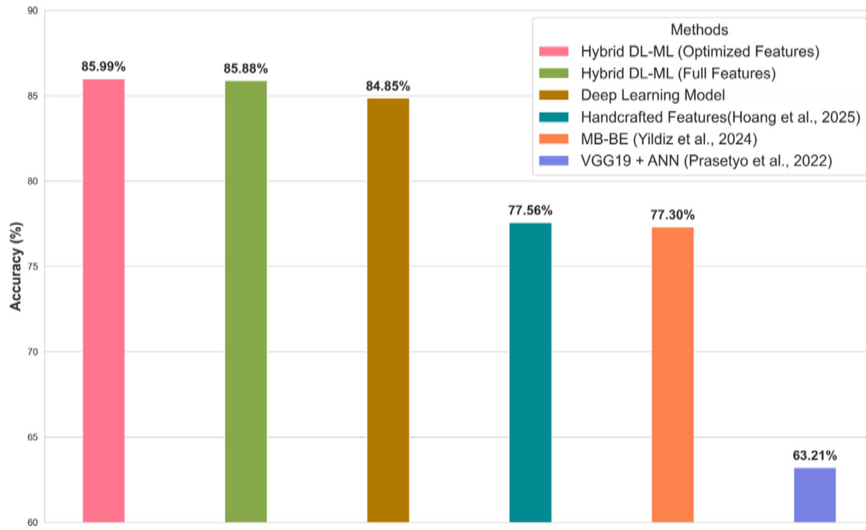


Fig. 8. Accuracy comparison of the proposed strategies with previous studies on the FFE dataset.

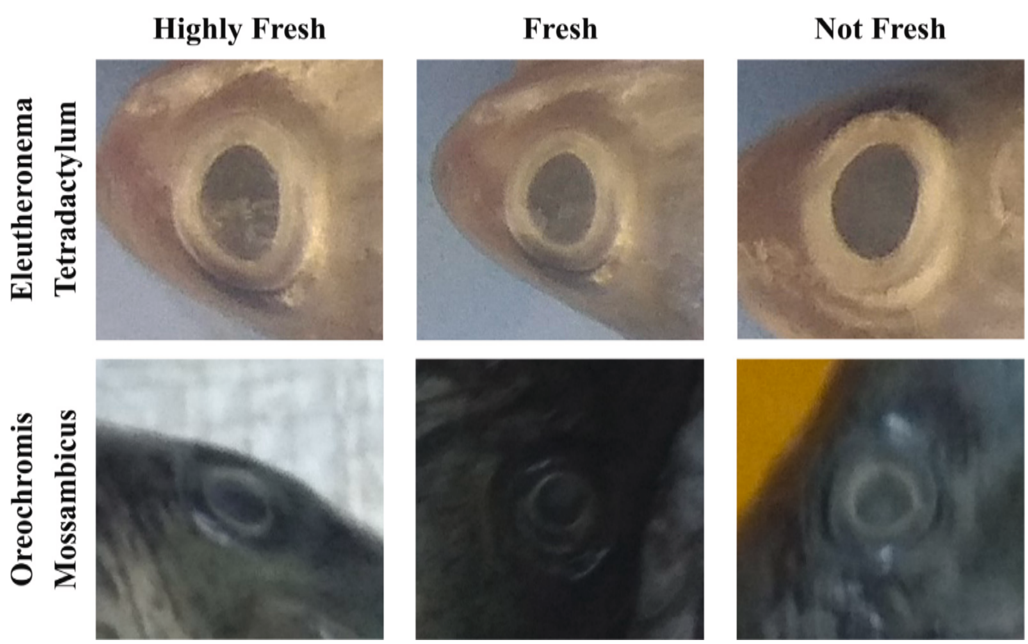


Fig. 9. Examples of ambiguous or misclassified fish-eye images from the FFE dataset (*Highly fresh*, *Fresh*, and *Not fresh*) for *Eleutheronema tetradactylum* and *Oreochromis mossambicus*.

Despite the promising results, the proposed framework is constrained by the inherent limitations of the FFE dataset, which comprises only 4390 images. This relatively small scale and idealized setup may limit the model's robustness to real-world variations, such as diverse lighting conditions, species diversity, or environmental noise encountered in industrial seafood processing environments. Fig. 9 illustrates representative ambiguous samples in which the visual differences between freshness classes are extremely subtle, making it challenging to distinguish class boundaries even for human observers.

In addition to dataset-related constraints, several methodological limitations should be noted. The three-stage, decoupled design increases implementation complexity, as each step – fine-tuning, feature extraction, feature selection, and classifier training – requires separate setup and optimization. While this modularity enhances interpretability and debugging flexibility, it also demands greater technical and computational resources, potentially hindering deployment in real-time or resource-constrained environments. Another limitation involves the exclusive use of deep features extracted from the Global Average Pooling (GAP), without incorporating handcrafted features. Deep features effectively capture high-level semantics but lack interpretable physical attributes. For instance, handcrafted color features (e.g., HSV histograms) could characterize eye redness, while texture descriptors (e.g., gray-level co-occurrence matrices) could quantify lens opacity. Combining both through feature fusion could yield a more comprehensive representation, integrating abstract contextual information with fine-grained physical cues.

Building upon these limitations, future research will focus on several concrete directions to extend the present study. First, developing hybrid representations that integrate deep features with handcrafted descriptors, such as HSV color histograms and texture-based opacity measures, could enable the joint capture of high-level semantic information and physically interpretable visual cues related to fish eye freshness. Second, validating the proposed framework on larger and more diverse fish eye freshness datasets, as well as across independent food-quality imaging benchmarks, will enable cross-dataset validation and a more rigorous assessment of its robustness and generalization potential under varying lighting conditions, fish species, and acquisition environments. Finally, further optimization for low-latency deployment, including integration into mobile or embedded systems, represents a promising avenue for supporting time-sensitive and real-time seafood quality inspection applications.

6. Conclusion

This study introduced a unified three-stage framework for optimizing deep visual representations in fish eye freshness assessment on the challenging FFE dataset. By fine-tuning state-of-the-art vision backbones, extracting multi-level features, applying classical ML classifiers, and performing embedded feature selection, the best configuration–Swin-Tiny features combined with an RF classifier and LGBM-based selection achieved an accuracy of 85.99%, outperforming prior studies on the same dataset by 8.69–22.78%. The results highlight the effectiveness of integrating vision transformers with boosting-based feature optimization for improved accuracy, interpretability, and efficiency in visual food quality tasks. Practically, this approach offers a scalable, objective tool for near real-time quality control in the seafood industry, helping minimize economic losses and enhance food safety. While limited by the FFE dataset's specific characteristics (e.g., eight species and uncontrolled mobile imaging), the flexible framework is generalizable to broader food imaging applications. Future work will validate it on diverse datasets, explore multimodal integration, and develop lightweight models for mobile deployment.

CRediT authorship contribution statement

Phi-Hung Hoang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Nam-Thuan Trinh:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Van-Manh Tran:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Thi-Thu-Hong Phan:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, Grammarly and generative large language models were employed to improve grammar and refine wording for clarity. The authors subsequently reviewed and revised the content as necessary and take full responsibility for the final version of the manuscript.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The dataset used in this study is available in the Mendeley Data repository (Prasetyo et al., 2022a).

The source code is available at GitHub Repository.

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